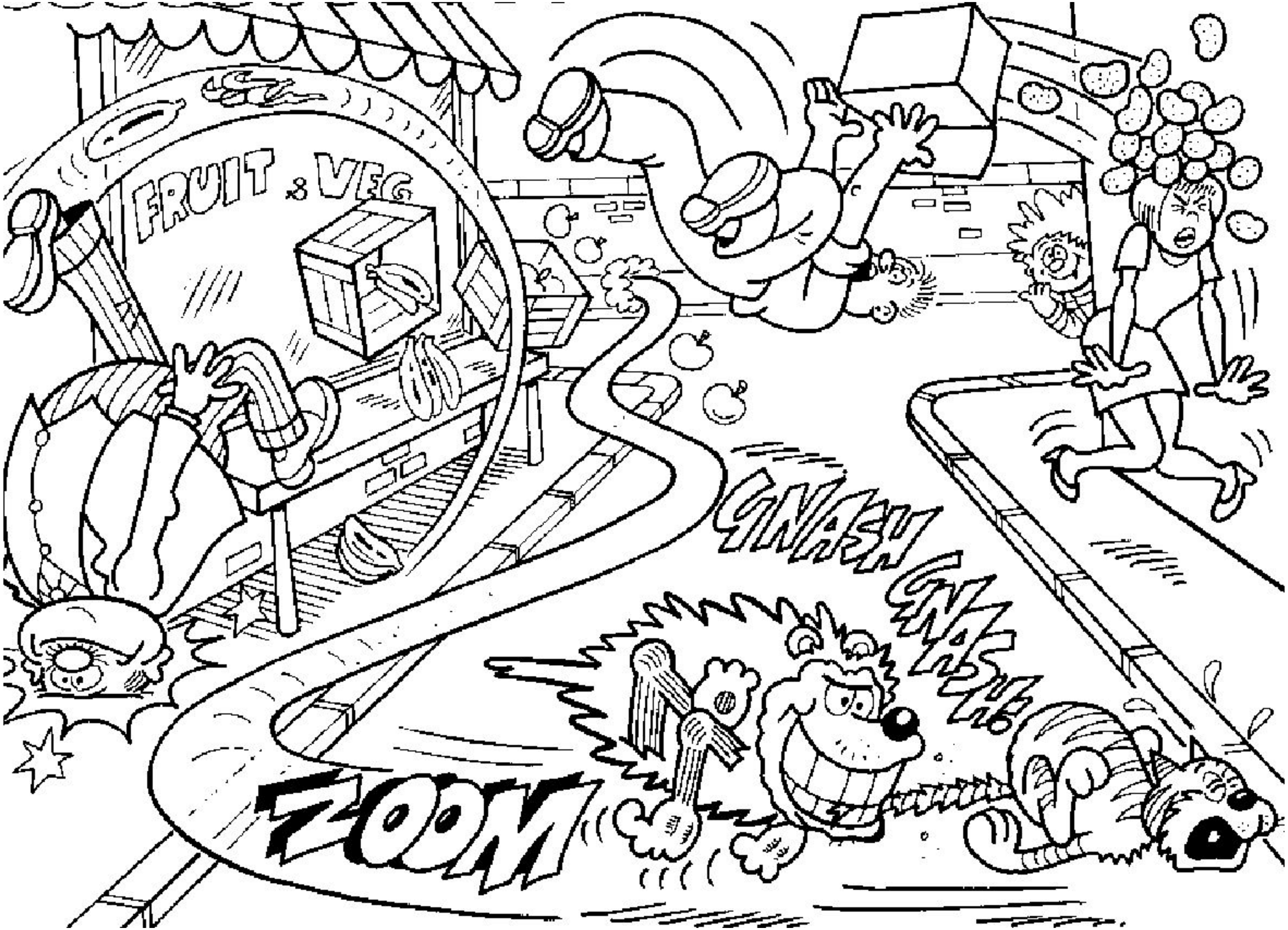
Chapter 3 Forces and Motion

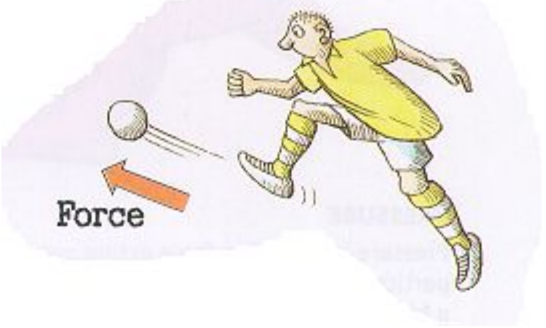
What are forces?


FRUIT & VEG
GNASH
GNASH
GNASH!
ZOOM

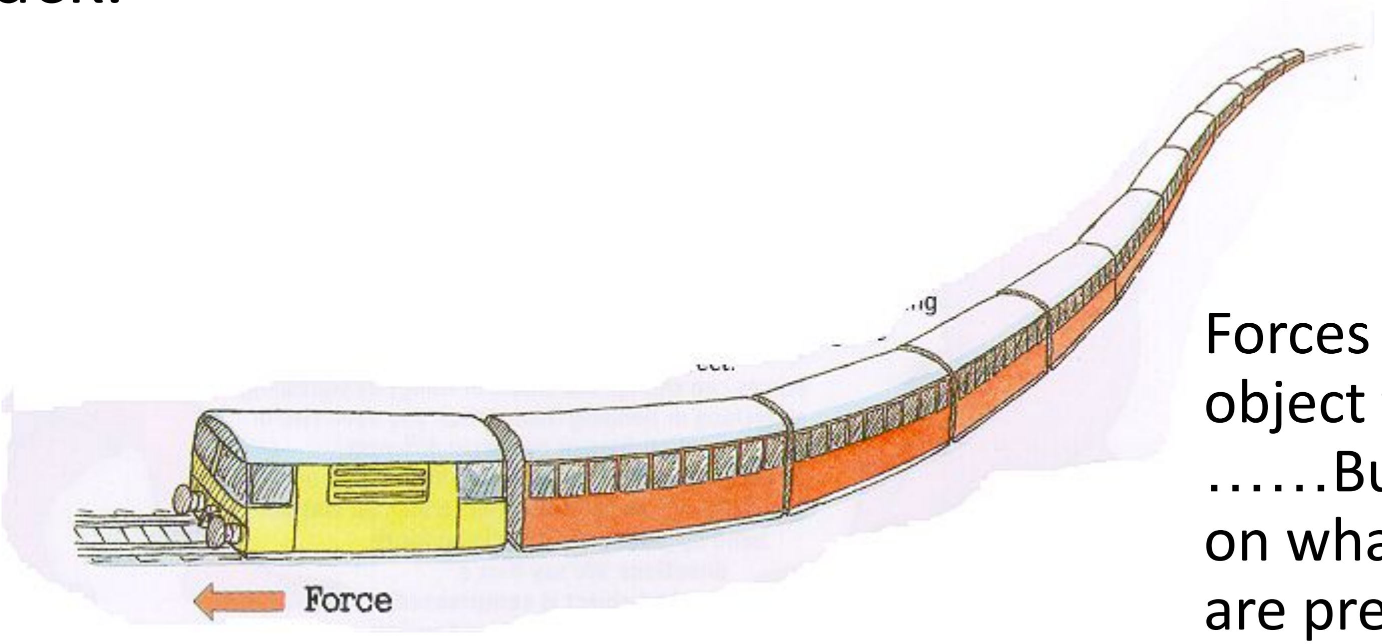
Success criteria

Lesson Title	Working towards	Working at	Working beyond
Recap: Forces & Motion (Y7)	I can identify different forces acting on an object.	I can describe what balanced and unbalanced forces are.	I can explain how unbalanced forces change an object's motion using force diagrams.
Speed	I can identify the correct units for speed, distance, and time.	I can use the speed = distance ÷ time formula to calculate speed, distance or time.	I can solve speed questions using different units and rearranged formulas.
Describing Movement	I can read values from a distance-time graph.	I can describe motion based on the shape of the graph.	I can draw and explain distance-time graphs from written data or scenarios.
Turning Forces	I can give examples of things that turn when a force is applied.	I can use the moment formula to calculate turning forces.	I can explain how balanced and unbalanced moments affect turning.
Pressure between solids	I can state that heavier or larger objects create more pressure.	I can calculate pressure using the correct formula and units.	I can compare pressure in different situations using scientific reasoning.
Pressure in liquids and gases	I can describe how pressure changes with depth in liquids and altitude in gases.	I can explain what causes pressure in liquids and gases using particles.	I can calculate liquid pressure using the $h\rho g$ formula and explain examples.
Particles on the move	I can define diffusion.	I can describe how temperature and concentration affect diffusion.	I can explain diffusion using the particle model and apply it to real examples.

What is a force?



- Forces are what makes things move.
- Forces are measured in <u>Newtons</u> (N)
- When you kick a football your foot applies a force to the ball.
- The engine of a train uses a pulling force to make the carriages move along the track.



Forces <u>may</u> cause an object to move ……But that depends on what other forces are present!

Forces are nearly always PUSHES or PULLS (or twists)

The greatest tennis rally ever!

How many different examples of forces can you give from the video?

A force can cause an object to do three things:

1. **A force can change an object's di__________.**

2. **A force can change an object's sp__________.**

   (we call this acceleration)

3. **A force can change an objects sh_______ and si__________.**

A force can cause an object to do three things:

1. A force can change an object's direction.

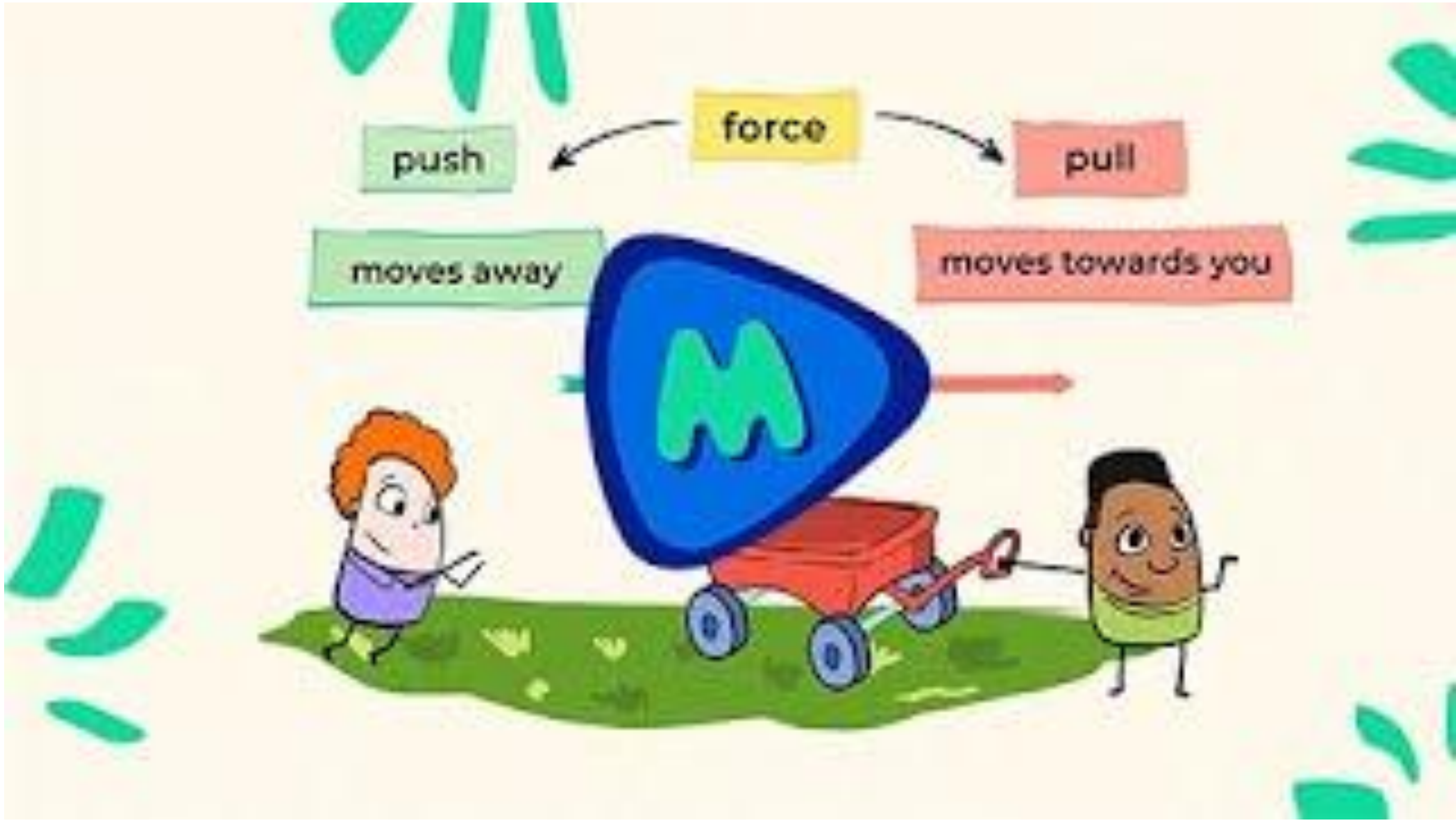
2. A force can change an object's speed.

(we call this acceleration)

3. A force can change an objects shape and size.

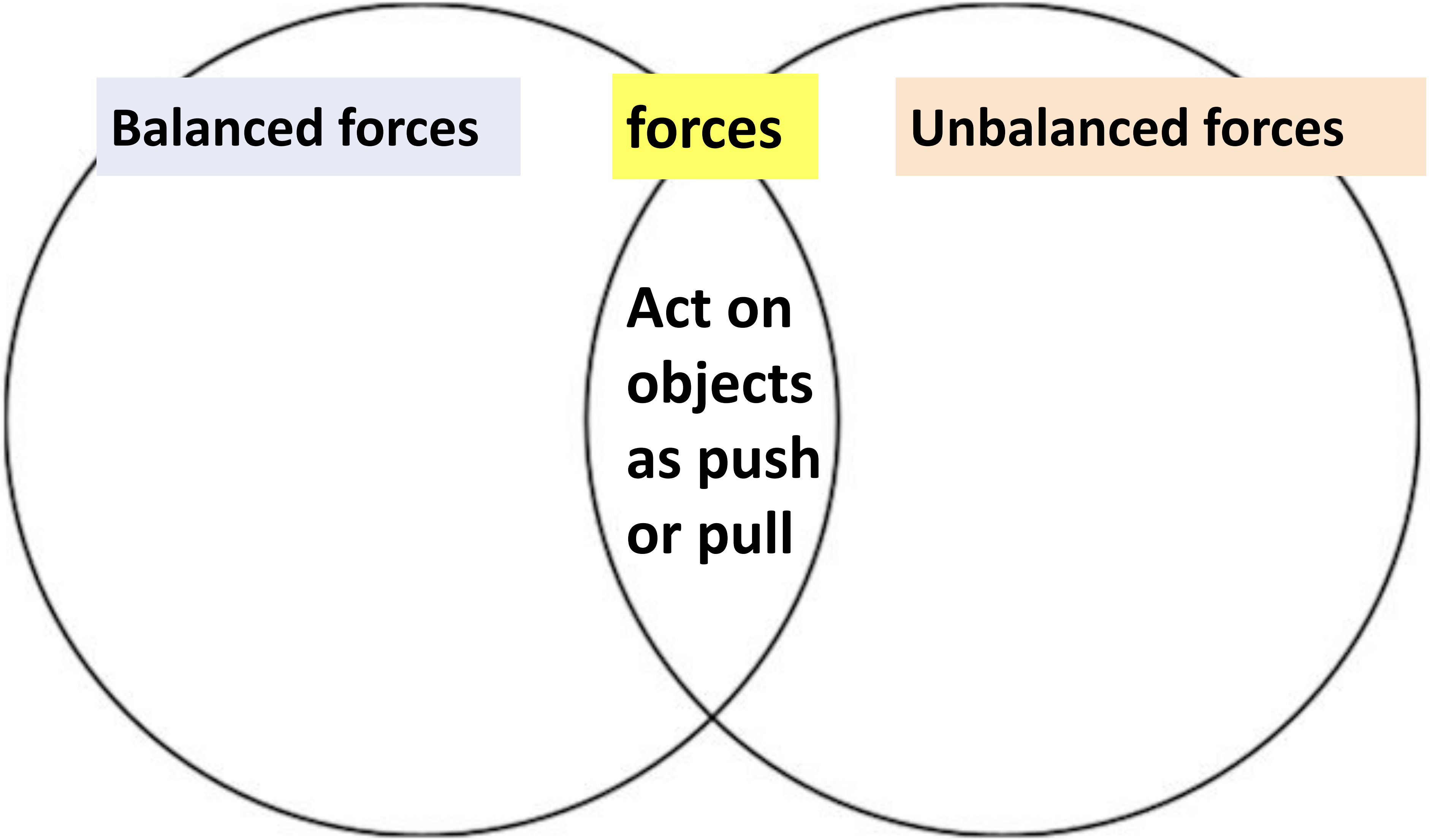
Balanced vs Unbalanced forces

Watch the video and answer:



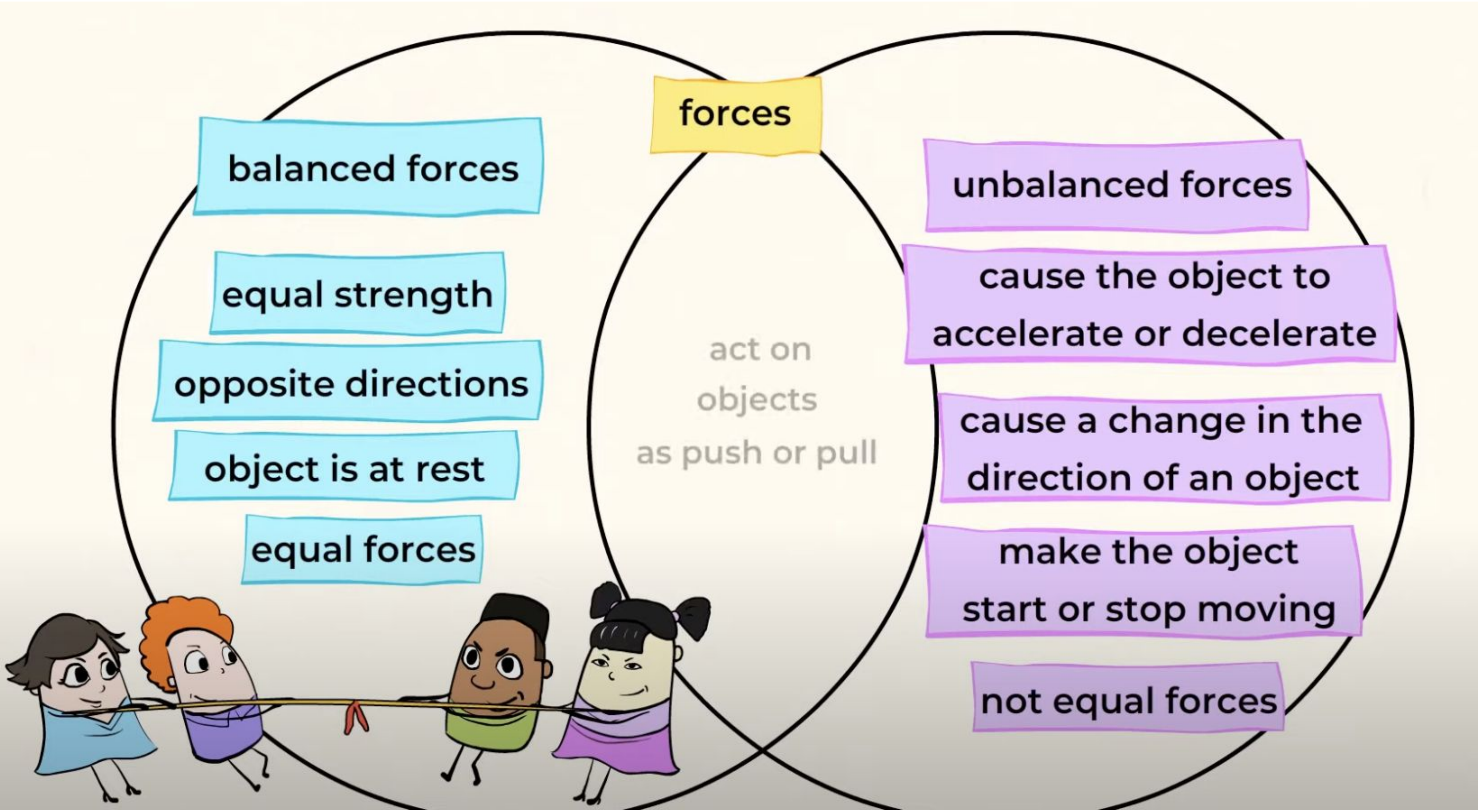
What is a balanced force?

What is an unbalanced force?


Balanced forces
forces
Unbalanced forces
Act on
objects
as push
or pull

Create a Venn diagram including the following statements

- Equal strength
- Cause a change in the direction of an object
- Equal forces
- Not equal forces
- Make the object start or stop moving
- Object is at rest
- Cause the object to accelerate or decelerate
- Opposite directions
- Act on objects as push or pull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